

TIMOTEJ GAŠPAR, PhD

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PhD in Robotics with a robust background in both academia and industry. Experienced in developing software and leading software development teams. An extremely quick learner and passionate about gaining new knowledge, both technical and soft skills. Strong sense of product oriented development with the ability to focus on creating tools for users, both internal and external. Track record of priority-driven project management through development and deployment of robot cells at client sites. Very detail-oriented, focused on adding value and driven by shipping good products.

Expertise: Robot software development, ROS/ROS 2, Motion planning, Robot control, Simulation, Software architecture, Project coordination, Team leadership, Technical documentation, Scrum master

Programming and development tools: Python, Linux, ROS/ROS 2, MoveIt, Isaac Sim, MATLAB, Docker, Git, CI/CD, C++

Work organization: Figma, GitHub, Linear, Notion, Atlassian, Scrum

Other trainings: Professional Scrum Master I, Professional Scrum Facilitation Skills, Python Asyncio

Languages: Slovenian, English, Italian

PROFESSIONAL EXPERIENCE

Sunrise Robotics, Europe – www.sunriserobotics.co

Senior robot software developer (remote)

2024 - 2025 (May)

- Contributed to early-stage software architecture, hardware design decisions, and development prioritization strategies as part of the founding engineering team.
- Designed and implemented the robot motion stack, including point-to-point trajectory generation (joint and Cartesian space) based on academic methods; integrated `ros2_control` and deployed the system on a PREEMPT_RT-patched Linux environment.
- Developed and maintained internal ROS 2 drivers for ABB GOFA series robots.
- Led the setup of the Isaac Sim-based simulation framework; created and optimized USD scenes and robot assets from CAD data, including mesh simplification, collider setup, and joint configuration.
- Built a ROS 2 driver for Siemens PLC signal integration using the Snap7 Python library.
- Shaped and co-developed the adoption of behavior trees for top-level robot cell process control.
- Explored integration of Gaussian Splatting volume rendering into the product stack; created visual assets and animations combining traditional 3D elements with splatted data.

The company ceased operations

Head of Software Development (remote)

2023 – 2024

- Led the development of the software stack for a reconfigurable industrial robot cell.
- Maintained and expanded high-quality software and hardware documentation (in English).
- Authored technical specifications for single pieces of software as well as complex robotic solutions (in English).
- Led reorganization of software and hardware development teams into Scrum teams, assuming the role of Scrum Master.
- Coordinated cross-functional teams (hardware, software, AI) for industrial robot cell development.
- Deployed updates/new features to remote robot cells at client sites.
- Set up a test coverage system and metrics for the proprietary Python software stack.
- Worked closely with the AI team and integrated AI perception into proprietary robotic software stack.
- Engineered a system for automated bill of materials compilation.
- Shaped company's mid and long-term strategic vision in collaboration with CEO/CTO.
- Served as the primary point of contact for delivering technical expertise on proprietary software stack, contributing to informed decision-making and ensuring alignment between technical capabilities and organizational objectives.

Head of Robotics Development (remote)

2021 – 2023

- Led a team of 5 developers and coded proprietary ROS-based robot framework (<https://github.com/flx-robotics/koco>).
- Led the deployment of 8 industrial robotic cells across different factories.
- Defined and co-authored the software architecture for a reconfigurable industrial robot cell.
- Defined coding conventions and implemented CI/CD processes, ensuring deployable and thoroughly tested code with pipelines that built and pushed Docker images.
- Engineered a system that stores production-critical information into a MongoDB database.
- Developed digital-twin system and principles for simulating robot systems for quicker development of both low- and top-level applications.
- Integrated MoveIt for collision-free path planning for industrial applications.
- Developed a robust system utilizing MODBUS protocol to seamlessly transmit real-time robot cell data to the Manufacturing Execution System (MES).
- Authored many proprietary ROS drivers in Python for industrial devices (e.g. robot drivers, quality assurance devices, etc.).
- Established robust documentation and peer review processes, ensuring code clarity and quality (in English).
- Empowered and mentored my team members towards clear and respectful communication.

Faculty of electrical engineering, University of Ljubljana

Teaching Assistant (part time)

2021 – 2023

- Organized and taught at the first and second (2021 & 2022) ROS workshop for students (<https://www.fe-ros.si/> & <https://ros-fe.github.io/fe-ros-docs>).
- Conducted Linux fundamentals workshops, providing essential skills to students.

“Jožef Stefan” Institute, Ljubljana, Slovenia

Robotics researcher

2015 – 2020

- Conducted extensive research on fast setup and reconfiguration of robot cells, contributing to EU projects (www.reconcell.eu, www.reconcycle.eu).
- Developed a hard-real-time robot controller for UR10 robots enabling complex trajectory planning algorithms and feedback control including force control through a force/torque sensor.
- Developed a framework with a GUI for kinesthetic teaching of robot motions (https://github.com/tgaspar/helping_hand).
- Developed software architecture in ROS, designed hardware solutions, and optimized layouts.
- Published multiple research papers, showcasing expertise in robotics and automation.

EDUCATION

Faculty of Electrical Engineering, University of Ljubljana

Doctorate of Electrical engineering (Robotics)

2015 – 2020

Masters degree of Electrical engineering (Robotics)

2010 – 2013

Bachelor degree in Electrical engineering

2010 – 2013

PUBLICATIONS IN SCIENTIFIC JOURNALS

- Gašpar, Timotej, Igor Kovač, and Aleš Ude. "Optimal layout and reconfiguration of a fixturing system constructed from passive Stewart platforms." *Journal of Manufacturing Systems* 60 (2021): 226-238.
- Gašpar, Timotej, et al. "Smart hardware integration with advanced robot programming technologies for efficient reconfiguration of robot workcells." *Robotics and Computer-Integrated Manufacturing* 66 (2020): 101979.
- Gašpar, Timotej, et al. "Skill learning and action recognition by arc-length dynamic movement primitives." *Robotics and autonomous systems* 100 (2018): 225-235.

See my Google Scholar Profile for a full list: <https://tinyurl.com/2tcmj7t8>

REFERENCES

Aleš Ude, Head of department for robotics at the “Jožef Stefan” Institute, ales.ude@ijs.si

Robert Bevec, CTO at Airnamics, (contact provided upon request)

Christian Schlette, Professor at SDU Robotics, chs@mmi.sdu.dk

Tomaž Štolfa, CEO of Sunrise Robotics, tomaz@sunriserobotics.co